

Almost Perfect Segmentation Without Better Models: The Impact of Evaluation Metrics

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Abstract—Rigorous evaluation of medical image segmentation models is essential for assessing their clinical utility, yet the choice and interpretation of evaluation metrics is often overlooked. In this work, we investigate the hypothesis that almost perfect segmentation results can be achieved by changing the evaluation metric. Using a classical random forest-based brain segmentation pipeline, we analyse the behaviour of overlap, distance, volume, and classification-based metrics. Our results show that substantial improvements in commonly reported metrics can be obtained through simple geometric manipulations, metric selection, and reporting strategies. These findings highlight the risk of overstating segmentation performance and the importance of careful metric selection and transparent evaluation in medical image analysis.

Index Terms—medical image segmentation, evaluation metrics, metric sensitivity, performance inflation, reporting bias

I. INTRODUCTION

Medical image segmentation is a fundamental task in medical image analysis, enabling quantitative assessment of anatomical structures and supporting downstream clinical and research applications. As segmentation methods continue to be introduced to clinical workflows, reliable evaluation has become increasingly important to ensure that reported performance accurately reflects anatomical correctness rather than artefacts of metric choice or reporting strategies.

Segmentation metrics respond differently to segmentation errors and are sensitive to factors such as object size, class imbalance or outliers [1]. Metrics are commonly grouped into overlap-, distance-, volume-, and classification-based categories with additional specialised metrics proposed for specific anatomical or clinical requirements [1], [2]. Overlap-based metrics quantify voxel-wise agreement, distance-based metrics focus on boundary discrepancies, volume-based metrics assess size agreement, and classification-based metrics quantify label assignment performance. Each category captures distinct aspects of performance while exhibiting characteristic failure modes [1].

In this work, we test the hypothesis that almost perfect segmentation results can be achieved by modifying the evaluation metric rather than improving the underlying segmentation. Using a classical random forest-based brain segmentation pipeline, we analyse metric sensitivity, as well as different reporting strategies such as label aggregation and global metrics.

II. MATERIALS AND METHODS

A. Dataset

The dataset consists of 30 healthy subjects from the Human Connectome Project [3]. For each subject, MR images are provided, including T1-weighted and T2-weighted scans. The images are accompanied by a manually annotated ground truth segmentation, brain masks and affine transformations to the MNI-152 atlas space. The ground truth labels include: background, white matter, grey matter, hippocampus, amygdala and thalamus. All images are bias-field corrected and provided in native image space.

B. Segmentation Pipeline

1) *Pre-processing*: All images are transformed into a common atlas space using the provided affine transformations. Skull stripping is performed using the supplied brain masks to remove non-brain tissue. Intensity normalisation is applied independently to T1-weighted and T2-weighted images using z-score normalisation. Ground truth segmentations are registered to the atlas space using nearest-neighbour interpolation to preserve label integrity.

2) *Classifier*: Voxel-wise classification is performed using a random forest classifier. For each voxel, the feature vector includes atlas coordinates, raw T1- and T2-weighted intensities and gradient-based features. The classifier consists of an ensemble of decision trees trained via bootstrap aggregation. Final predictions are obtained by majority voting across trees, reducing variance.

3) *Post-processing*: The initial segmentation output is refined using morphological post-processing. Small connected components below a predefined size threshold are removed independently for each label and all segmentations are masked to the brain region to enforce anatomical plausibility. This intentionally simplistic post-processing enables controlled manipulation of segmentation geometry for metric sensitivity analysis.

4) *Segmentation Evaluation*: Segmentation performance is assessed by comparing predicted labels with the registered ground truth. Evaluation metrics include overlap-based metrics, volume-based metrics, distance-based metrics, and classification-based metrics. Metrics are computed on a per-label basis and aggregated across subjects.

C. Experimental Design

Three experiments are conducted to test the stated hypothesis. The first experiment analyses the sensitivity of segmentation metrics to controlled changes in the pipeline. The second experiment investigates whether overall evaluation metrics can conceal poor performance on individual anatomical structures. The third experiment examines whether reducing label granularity inflates reported performance on small anatomical structures.

1) *Metric Sensitivity Analysis* : To analyse metric sensitivity, firstly a series of post-hoc perturbations are applied to the segmentation output while keeping the underlying classifier unchanged.

Largest connected component filtering removes all but the largest connected component for each anatomical label. Distance-based component removal eliminates components that are spatially distant from the main structure of each label, using label-specific distance thresholds to reflect anatomical size differences. Boundary shrinkage applies a mild morphological erosion to each label, modifying boundary accuracy while preserving the overall spatial location. Morphological volume adjustment applies label-wise opening or closing operations to alter predicted volumes, bringing them closer to the ground truth volume.

In addition, a restricted-label model is trained that predicts only the background class. This intentionally anatomically meaningless segmentation serves as a stress test for classification-based metrics.

2) *Overall versus Per-label Metric Evaluation*: To compare overall and per-label evaluation, three additional random forest models are trained: one that balances the class weights, one weighted towards large anatomical structures, and one weighted towards small anatomical structures. Overall metrics are computed directly from the aggregated confusion matrix or distance distributions, rather than averaging per-label metrics to ensure mathematical consistency. The performance of these models is compared with that of the standard pipeline.

3) *Evaluation under Reduced Label Granularity*: To study the effect of label aggregation, the amygdala, hippocampus and thalamus, are grouped together into a single "small structures" label. The standard pipeline is then evaluated using three labels: white matter, grey matter and small structures. Performance under reduced granularity is compared to the original per-label evaluation.

III. RESULTS

A. Metric Sensitivity Analysis

We begin by analysing the sensitivity of commonly used segmentation metrics to post-hoc perturbations affecting distance properties of the segmentation. Table I summarises the changes in Hausdorff distance, a distance-based metric, and Dice coefficient, an overlap-based metric, across anatomical structures. Across the distance-based perturbations, Dice scores change only marginally, indicating limited sensitivity of overlap-based metrics to these manipulations. In contrast,

Hausdorff distance exhibits large and often directionally inconsistent changes, demonstrating strong sensitivity to spatial changes.

TABLE I: Difference in overlap- and distance-based metrics induced by three post-hoc segmentation perturbations

Label	Largest CC		Remove distant comp.		Boundary shrink	
	Δ Dice	Δ HD	Δ Dice	Δ HD	Δ Dice	Δ HD
Amygdala	+0.003	+0.029	+0.003	+0.029	-0.028	-4.291
Grey matter	+0.001	-0.104	+0.001	-0.104	-0.477	+11.214
Hippocampus	+0.007	-0.498	+0.005	-0.05	-0.046	-6.063
Thalamus	+0.012	-9.459	+0.012	-9.459	+0.009	-11.302
White matter	-0.003	+2.231	-0.003	+2.231	-0.412	+11.523

Largest connected component filtering and distance-based component removal produce nearly identical effects. As shown in Figure 1 and Figure 2, both methods substantially reduce Hausdorff distance for the thalamus, a compact structure, while increasing Hausdorff distance for white matter, a large, diffuse structure. In contrast, Dice scores remain largely unchanged as reported in Table I, confirming that enforcing spatial connectivity primarily affects extreme surface distances rather than overall overlap quality. This highlights that distance-based metrics are often influenced by isolated spatial outliers rather than global segmentation correctness.

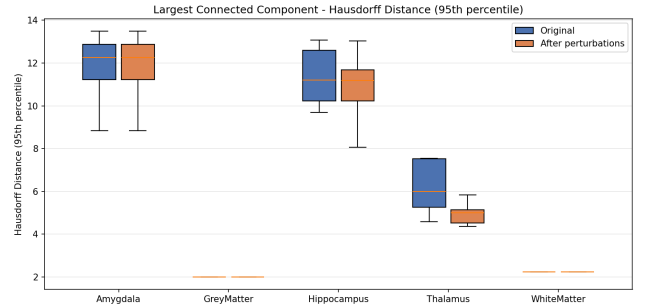


Fig. 1: Largest connected component perturbation compared to the original segmentation using Hausdorff distance

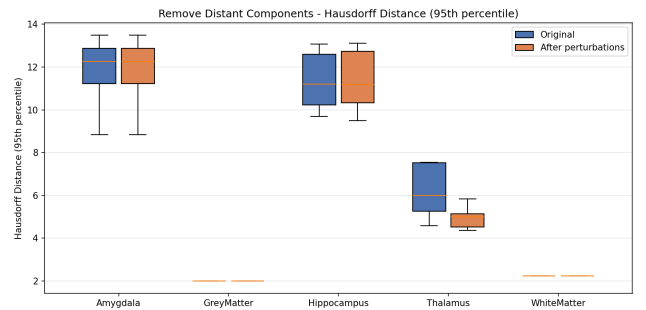


Fig. 2: Distance-based component removal compared to the original segmentation using Hausdorff distance

Boundary shrinkage produces a more heterogeneous response. Figure 3 and Table I show that, Hausdorff distance increases tremendously for large anatomical structures suggesting that boundary shrinkage introduces surface discrepancies

in large structures. In contrast, small and compact structures such as the thalamus show a noticeable decrease in the Hausdorff distance. Although Dice scores decrease more than in component-removal methods, these changes remain small relative to the pronounced variation in Hausdorff distance. This highlights that distance-based metric behaviour is influenced by object size and morphology.

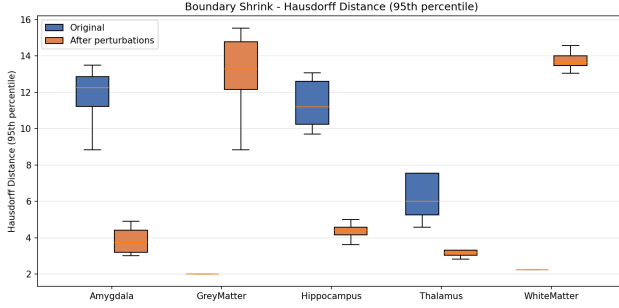


Fig. 3: Boundary shrinking perturbation compared to the original segmentation using Hausdorff distance

Secondly, the effect of morphological volume adjustment is examined. Label-wise opening or closing operations are applied to reduce volume discrepancies between predictions and ground truth. As shown in Figure 4, this leads to near-perfect volume similarity scores in the large anatomical structures and significant improvements in the smaller structures. These results demonstrate that volume-based metrics can be trivially inflated through post-processing operations that do not meaningfully improve spatial alignment or boundary accuracy.

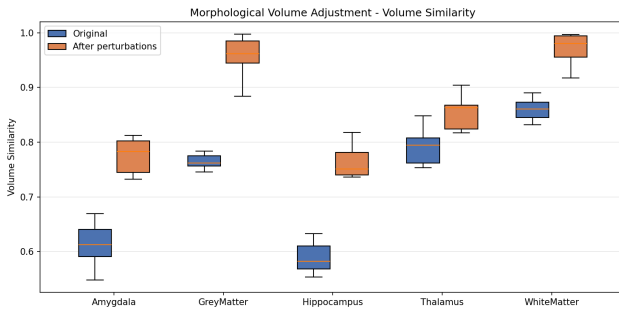
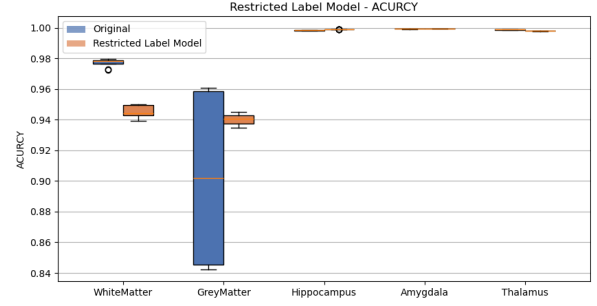
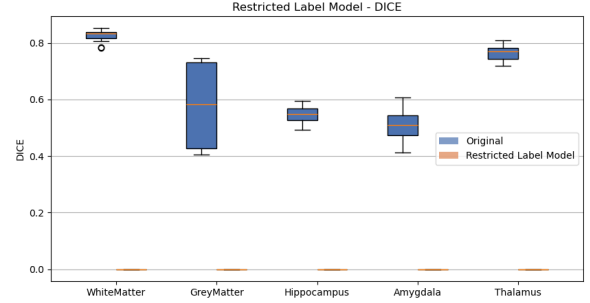


Fig. 4: Morphological changes perturbation compared to the original segmentation using Volume Similarity

Finally, a restricted label model that assigns all voxels to the background class is evaluated to assess the behaviour of classification-based metrics under class imbalance. Despite producing no meaningful anatomical segmentation, this model, as shown in Figure 5, achieves high accuracy due to the dominance of background voxels, while Dice scores for all anatomical structures are zero. This illustrates how classification-based metrics can be inflated under class imbalance due to confusion-matrix based computation.



(a) Accuracy



(b) DICE score

Fig. 5: Evaluating the background restricted model compared to the standard model

B. Overall metrics vs Per-label metrics

Figure 6 compares overall evaluation and per-label evaluation for four differently weighted models using the Hausdorff distance metric. Overall evaluations, represented by dashed lines, appear similar across all models, masking substantial label-wise performance differences. The balanced-weight model exhibits poorer performance on large anatomical structures, characterised by higher medians and a greater interquartile range. In contrast, the model weighted towards large structures performs well on white matter and grey matter but shows reduced performance on smaller structures such as the hippocampus and thalamus. These divergent behaviours are obscured in the overall evaluation, demonstrating that aggregated metrics can conceal divergent behaviour across anatomical labels.

C. Aggregated Evaluation

Evaluation using reduced label granularity results in improved median performance in comparison to the worst performing small anatomical structure, as seen in Figure 7. Using aggregated labels can therefore obscure poor performance on individual small structures. Consequently, this gives a misleading impression of robustness.

IV. DISCUSSION

The hypothesis tested was that almost perfect segmentation results can be achieved by modifying the evaluation metrics

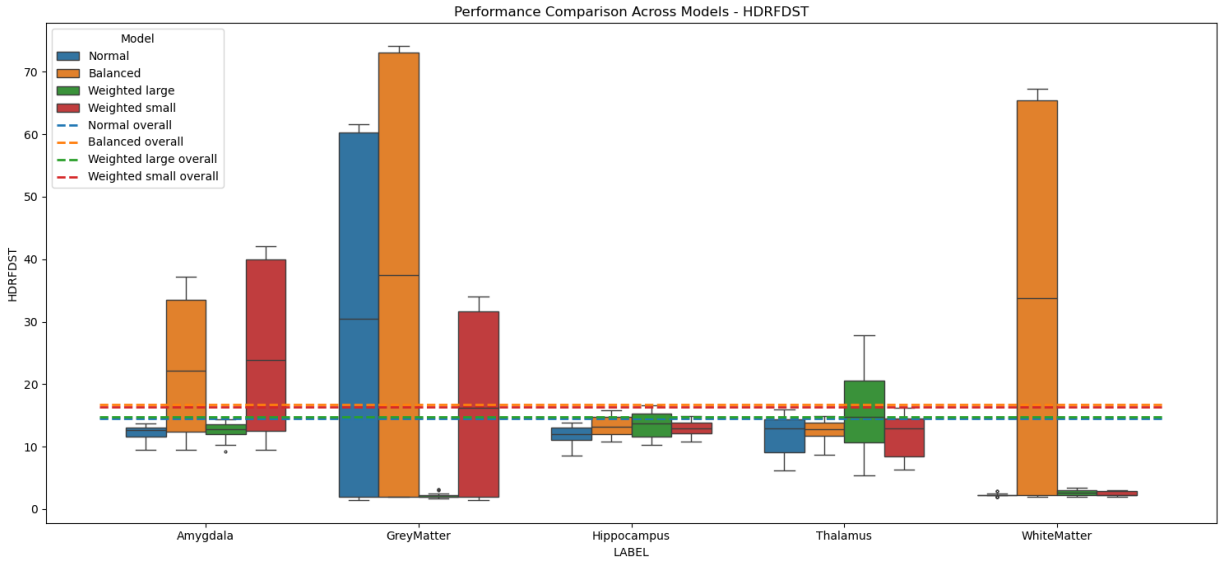


Fig. 6: Overall and Per-label evaluation of various weighted models using Hausdorff distance

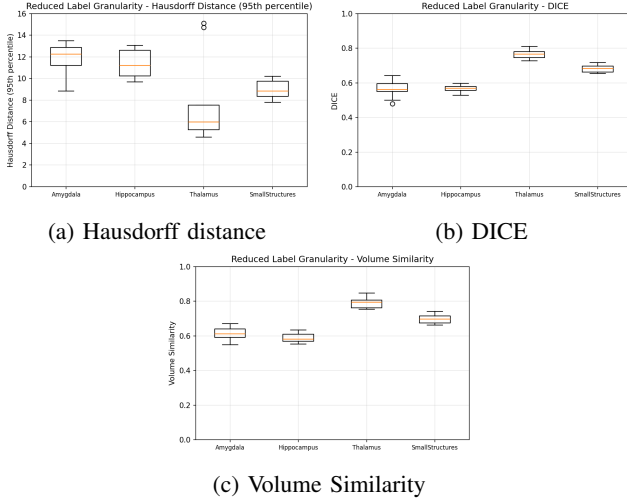


Fig. 7: Comparison of reduced label granularity evaluation and per-label evaluation on small anatomical structures

or reporting strategies rather than improving the underlying segmentation quality. The experimental results support this hypothesis by demonstrating that reported performance is highly dependent on metric choice, label aggregation, and post-processing operations, often without corresponding improvements in true anatomical accuracy. The metric sensitivity analysis shows that segmentation metrics are not equally robust to different post-hoc perturbations, highlighting how perceived performance can be strongly influenced by both the chosen metric and post-processing strategy. The metric sensitivity experiments also revealed that class imbalance can lead to highly inflated metrics that suggest near-perfect performance despite poor segmentation quality. Furthermore, comparisons between overall and per-label evaluation show

that aggregated metrics can hide large performance differences across anatomical structures. Similarly, reduced label granularity consistently improved median scores, masking poor performance on small structures.

Several limitations should be acknowledged. Firstly, this study relies on a classical machine learning pipeline based on a random forest model. While this choice enabled controlled experimentation with weighted models, deeper models have demonstrated state-of-the-art performance in medical image segmentation [4]. Secondly, the experiments focus on evaluation design rather than on improving the segmentation pipeline itself, which limits conclusions about real-world model development scenarios.

Despite these limitations, the findings of this work demonstrate the strong dependence of segmentation evaluation on metric choice and reporting strategy, and highlight the risk of overstating segmentation quality through inappropriate evaluation design. This work provides empirical evidence for the hypothesis and demonstrates how misleading conclusions can arise from inappropriate evaluation design.

V. CONCLUSION

This work demonstrates that reported segmentation performance is highly sensitive to evaluation metric choice, and reporting strategies such as label aggregation and global evaluation. Through controlled experiments, we showed that near perfect metric scores could be achieved without improving the segmentation quality. These results confirm the hypothesis that segmentation evaluation can be manipulated through metric selection alone. The findings highlight the necessity of transparent, label-wise reporting and the use of multiple different metrics when evaluating medical image segmentation models to ensure that reported performance reflects clinical and anatomical relevance while avoiding bias towards specific metric properties.

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